

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester II)
Scala for Data Science Practical

Practical - 8

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Class :SYIT	Batch:01
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Practical o8

Find the correlation between two lists of numbers. Implement the formula for Person correlation coefficient.

INPUT

```
object S009 extends App {  
  var num = 0.0  
  
  for (i <- 0 to x.length - 1) {  
    num = num + (x(i) - meanX) * (y(i) - meanY)  
  }  
  
  var denX = 0.0  
  var denY = 0.0  
  
  for (i <- 0 to x.length - 1) {  
    denX = denX + (x(i) - meanX) * (x(i) - meanX)  
  }  
  
  for (i <- 0 to y.length - 1) {  
    denY = denY + (y(i) - meanY) * (y(i) - meanY)  
  }  
  
  val den = math.sqrt(denX * denY)  
  val cor = num / den  
  println("Pearson Correlation Coefficient: " + cor)  
  
  println("S006 Sejal Deshmukh")  
}
```

OUTPUT

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...  
Pearson Correlation Coefficient: 1.0  
S006 Sejal Deshmukh  
  
Process finished with exit code 0
```